Muon decay rate analysis using Monte Carlo acceptance-rejection method for a muon bound to a nucleus

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***Abstract*—This document presents a study of the Monte Carlo acceptance-rejection method applied to the differential decay rate of the muon bound to a nucleus, including radiative corrections, with the goal of understanding sampling techniques used in High Energy Physics simulations, as well as the analysis of the radiative terms, its divergences and the overall range of its variables. The complete process has been throughly documented in this repository, including the implementation of the Acceptance-Rejection Method (ARM) in Python and the analysis of the radiative corrections. The results indicate a well-defined behavior corresponding with the article published by Czarnecki et al. in 2014, which is the main reference for this work.**

I. Introduction

The Monte-Carlo method is a tool used widely in computer simulations in experimental physics. Here, I will study this mathematical and computational technique to enhance my understanding of simulations in High Energy Physics (HEP), the main reference for this learning process is chapter 13 from the book *Probability for Physicists* [1] specifically used for the Monte Carlo method, and the article *Michel decay spectrum for a muon bound to a nucleus* [2] which is the main reference for the analysis of the radiative corrections in muon decay.

The Monte-Carlo method considered in this work is the ARM, a technique used to generate random samples from a probability distribution that may be difficult to sample from directly, the method works by generating random samples from a proposal distribution and accepting or rejecting those samples according to a criterion that guarantees that the accepted samples follow the desired target distribution [3].

The main reference of this work is the article *Michel decay spectrum for a muon bound to a nucleus* [2], which provides a detailed analysis of the radiative corrections to the Michel decay spectrum, including the derivation of the differential decay rate and the implementation of the radiative corrections, which is the main focus of this work. The article also provides a comparison of the theoretical predictions with experimental data, which serves as a benchmark for the Monte Carlo simulations developed in this work, although the experimental data is not included in this repository, the theoretical predictions and graphical representations are used as a reference for the analysis of the radiative corrections.

II. Methodology

A. Acceptance-Rejection mathematics

This will be the method used in the Monte Carlo analysis developed in this repository, since the probability density function (PDF) of the decay spectrum is already known, it can be directly employed as the target distribution in the ARM.

B. Target distribution, muon decay rate and radiative corrections

HERE AN EXPLANATION MORE THOROUGH OF THE TARGET DISTRIBUTION AND THE RADIATIVE CORRECTIONS, INCLUDING THE EQUATIONS AND DEFINITIONS. EXPLANATION HERE OF THE PAPER!!!

In our case, the target distribution corresponds to the differential decay rate $\frac{d\Gamma}{dx}$, where x is the normalized electron energy defined as

$$x = \frac{2E_e}{m_\mu}, \qquad 0 < x \leq 1,$$

with E_e the electron energy and m_μ the muon mass.

Neglecting terms of order m_e^2/m_μ^2 and higher-order radiative corrections $\mathcal{O}(\alpha^2)$, the differential decay rate for free muon decay is given by

$$\frac{d\Gamma^{\text{free}}}{dx} = \frac{G_F^2 m_\mu^5}{192\pi^3}\, x^2 \left(6 - 4x + \frac{\alpha}{\pi} f(x)\right) \tag{1}$$

where G_F is the Fermi constant and $\alpha \approx \frac{1}{137}$ is the fine-structure constant, having in mind that the constantes used in the Python code come from the library `scipy.constants.physical_constants`, which in escense comes from the CODATA 2022 NIST database [4].

The function $f(x)$ represents the radiative corrections to the electron energy spectrum and is explicitly written as:

$$f(x) = \left[\frac{5}{3x^2} + \frac{16x}{3} + \frac{4}{x} + (12 - 8x)\ln\left(\frac{1}{x} - 1\right) - 8\right]\ln\left(\frac{m_\mu}{m_e}\right)$$
$$+ (6 - 4x)\Bigg[2\,\mathrm{Li}_2(x) - 2\ln^2(x) + \ln(x)$$
$$+ \ln(1-x)\left(3\ln(x) - \frac{1}{x} - 1\right) - \frac{\pi^2}{3} - 2\Bigg]$$
$$+ \frac{1-x}{3x^2}\left[34x^2 + (5 - 34x^2 + 17x)\ln(x) - 22x\right]$$
$$+ 6(1-x)\ln(x). \tag{2}$$

This expression is implemented directly in the Monte Carlo framework and used as the target probability density in the acceptance rejection algorithm for generating electron energy spectra from muon decay [2].

Comportamiento límite de las funciones en el Monte Carlo

Sea la tasa diferencial de decaimiento

$$\frac{d\Gamma}{dx} \equiv \_dGamma\_dx\_differential\_decay\_rate(x),$$

la cual ya incluye el factor global x^2 en su definición interna.

La parte radiativa se modela mediante

$$f(x) \equiv \_f\_radiative\_corrections(x),$$

que contiene términos singulares como $\frac{1}{x}$, $\frac{1}{x^2}$ y $\ln(x)$.

Para analizar su contribución efectiva, se considera el producto

$$x^2 \cdot f(x).$$

Límites en los bordes

- Cuando $x \to 0^+$:

$$\lim_{x\to 0^+} x^2 f(x) \quad \text{es finito.}$$

- Cuando $x \to 1^-$:

$$\lim_{x\to 1^-} x^2 f(x) \quad \text{es también finito.}$$

Conclusión

El producto $x^2 f(x)$ asegura que la parte radiativa aporta una contribución bien definida en ambos extremos del dominio $x \in (0, 1)$.

References

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